



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 • STANDARD 6.GR.1

# Area of Regular Polygons

Lesson 5-4 • Teacher Copy

**Name:** Type your name      **Date:** Today's date      **Class:** Class period

---



## TODAY'S OBJECTIVES

**Content Objective:** I can find the area of a regular polygon by decomposing it into triangles.

**Language Objective:** I can explain my steps using the words regular polygon, decompose, triangle, and composite.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Regular Polygon

Decompose

Triangle

Composite

Formula



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

A polygon with all sides and all angles equal is a .

2

To break a shape into smaller shapes is to  it.

3

A three-sided polygon is a .

4

Made of two or more shapes combined, a figure is .

5

A math rule written with symbols is a .

## Reflect — Exit Ticket

A regular pentagon is decomposed into 5 triangles from its center. Each triangle has a base of 7 inches and a height of 4.8 inches. What is the total area of the pentagon?

- A. 84 sq in
- B. 16.8 sq in
- C. 168 sq in
- D. 84 in

Your answer:

---

---

---

---

---

## Answer Key & Teacher Guide

1. **Notes 1:** Regular Polygon
2. **Notes 2:** Decompose
3. **Notes 3:** Triangle
4. **Notes 4:** Composite
5. **Notes 5:** Formula
6. **Watch for this mistake:** *A common mistake in Area of Regular Polygons is skipping the key idea: "A regular polygon splits into the same number of triangles as it has sides, so total area = (one triangle's area)  $\times$  (number of sides)." — always check your work against this rule before you submit.*
7. **Try It 1:** A.  $60 \text{ in}^2$  — *One triangle =  $\frac{1}{2} \times 4 \times 6 = 12 \text{ in}^2$ . A pentagon has 5 triangles, so total =  $5 \times 12 = 60 \text{ in}^2$ .*
8. **Try It 2:** A.  $120 \text{ sq ft}$  — *An octagon has 8 sides, so it splits into 8 triangles. One triangle =  $\frac{1}{2} \times 6 \times 5 = 15 \text{ sq ft}$ . Total =  $8 \times 15 = 120 \text{ sq ft}$ .*
9. **Exit Ticket:** A.  $84 \text{ sq in}$  — *One triangle:  $A = \frac{1}{2} \times 7 \times 4.8 = 16.8 \text{ sq in}$ . Total =  $5 \times 16.8 = 84 \text{ square inches}$ .*